ANSWER KEY & MARKING SCHEME SET 1

Chapter 9: **Symmetry** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 18	Lines of symmetry = $360 \div$ angle.
2	**(d)** 300	Multiply.
3	**(b)** A square	Only the square among these has 4.
4	**(d)** 60	The square fits after every multiple of 90°.
5	**(b)** 10	Add.
6	**(a)** 5	Count the letters with a fold line.
7	**(c)** 45	$360 \div$ spokes.
8	**(b)** 20	Lines of symmetry = $360 \div$ angle.
9	**(d)** 225	Multiply.
10	**(d)** hexagon	6 lines mean 6 sides.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Sides $= 360 \div 40 = 9$ (1); Order $= 9$ (1).	1+1
12	square: 4 (1); regular octagon: 8; total 12 (1).	1+1
13	Angle between neighbouring vertices $= 360 \div 6 = 60^\circ$ (1); Turn $= 2 \times 60 = 120^\circ$ (1).	1+1
14	Sides $= 360 \div 36 = 10$ (1); Order $= 10$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Angle $= 360 \div 5 = 72^\circ$ (1); After 3 turns: 216° (1); Turns for full rotation $= 5$ (1).	1+1+1
16	H: order 2, lines 2; A: order 1, lines 1; T: order 1, lines 1 (1); Compare each pair (1); Counting complete (1).	1+1+1
17	Angle between marks $= 360 \div 12 = 30^\circ$ (1); Lines of symmetry $= 12$ for a regular arrangement (1); Turn to mark 5: 150° (1).	1+1+1
	OR: Angle $= 360 \div 6 = 60^\circ$ (1); After 4 turns: 240° (1); Turns for full rotation $= 6$ (1).	1+1+1
18	O: order 2, lines 2; X: order 2, lines 2; Z: order 2, lines 0 (1); Compare each pair (1); Counting complete (1).	1+1+1
	OR: Angle between marks $= 360 \div 8 = 45^\circ$ (1); Lines of symmetry $= 8$ for a regular arrangement (1); Turn to mark 3: 135° (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(a) 5 (1); (b) 72° (1); (c) 5 (1); (d) 144° is 2 turns so it leaves the pattern unchanged (2).	1+1+1+2
	OR: square: 90° (1); regular octagon: 45° (1); Larger angle: square; difference $= 45^\circ$ (1); Total lines $= 4 + 8 = 12$ (2).	1+1+1+2
20	(a) 6 (1); (b) 60° (1); (c) 6 (1); (d) 180° is 3 turns so it leaves the pattern unchanged (2).	1+1+1+2
	OR: equilateral triangle: 120° (1); regular hexagon: 60° (1); Larger angle: equilateral triangle; difference $= 60^\circ$ (1); Total lines $= 3 + 6 = 9$ (2).	1+1+1+2

Mistakes to avoid:

- Saying a rectangle has 4 lines of symmetry (diagonals are not lines of symmetry); a non-square rectangle has only 2.
- Giving a parallelogram 2 lines of symmetry; it has none, though it has rotational symmetry of order 2.
- Mixing up the order of rotation with the smallest angle: angle $= 360^\circ \div$ order.
- Saying a figure with order 1 has rotational symmetry; order 1 means only the trivial full turn.
- Drawing the mirror image at the wrong distance from the mirror line or on the same side of it.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 9: **Symmetry** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 40	The square fits after every multiple of $90°$.
2	**(d)** 8	Add.
3	**(b)** 2	Count the letters with a fold line.
4	**(c)** 40	$360 \div$ spokes.
5	**(a)** 24	Lines of symmetry = $360 \div$ angle.
6	**(d)** 216	Multiply.
7	**(a)** equilateral triangle	Equilateral triangle.
8	**(a)** 75	The square fits after every multiple of $90°$.
9	**(d)** 12	Add.
10	**(a)** 3	Count the letters with a fold line.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	equilateral triangle: 3 (1); regular hexagon: 6; total 9 (1).	1+1
12	Angle between neighbouring vertices $= 360 \div 8 = 45°$ (1); Turn $= 3 \times 45 = 135°$ (1).	1+1
13	Sides $= 360 \div 30 = 12$ (1); Order $= 12$ (1).	1+1
14	regular pentagon: 5 (1); square: 4; total 9 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Angle $= 360 \div 8 = 45°$ (1); After 5 turns: $225°$ (1); Turns for full rotation $= 8$ (1).	1+1+1
16	T: order 1, lines 1; I: order 2, lines 2; N: order 2, lines 0 (1); Compare each pair (1); Counting complete (1).	1+1+1
17	Angle between marks $= 360 \div 10 = 36°$ (1); Lines of symmetry $= 10$ for a regular arrangement (1); Turn to mark 4: $144°$ (1).	1+1+1
	OR: Angle $= 360 \div 9 = 40°$ (1); After 4 turns: $160°$ (1); Turns for full rotation $= 9$ (1).	1+1+1
18	M: order 1, lines 1; O: order 2, lines 2; S: order 2, lines 0 (1); Compare each pair (1); Counting complete (1).	1+1+1
	OR: Angle between marks $= 360 \div 6 = 60°$ (1); Lines of symmetry $= 6$ for a regular arrangement (1); Turn to mark 5: $300°$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(a) 8 (1); (b) $45°$ (1); (c) 8 (1); (d) $135°$ is 3 turns so it leaves the pattern unchanged (2).	1+1+1+2
	OR: regular pentagon: $72°$ (1); regular decagon: $36°$ (1); Larger angle: regular pentagon; difference $= 36°$ (1); Total lines $= 5 + 10 = 15$ (2).	1+1+1+2
20	(a) 9 (1); (b) $40°$ (1); (c) 9 (1); (d) $160°$ is 4 turns so it leaves the pattern unchanged (2).	1+1+1+2
	OR: square: $90°$ (1); regular hexagon: $60°$ (1); Larger angle: square; difference $= 30°$ (1); Total lines $= 4 + 6 = 10$ (2).	1+1+1+2

Mistakes to avoid:

- Saying a rectangle has 4 lines of symmetry (diagonals are not lines of symmetry); a non-square rectangle has only 2.
- Giving a parallelogram 2 lines of symmetry; it has none, though it has rotational symmetry of order 2.
- Mixing up the order of rotation with the smallest angle: angle $= 360° \div$ order.
- Saying a figure with order 1 has rotational symmetry; order 1 means only the trivial full turn.
- Drawing the mirror image at the wrong distance from the mirror line or on the same side of it.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 9: **Symmetry** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 36	$360 \div$ spokes.
2	**(d)** 30	Lines of symmetry = $360 \div$ angle.
3	**(c)** 160	Multiply.
4	**(d)** isosceles triangle	A turn of 360 degrees is the only turn that fits.
5	**(d)** 20	The square fits after every multiple of $90°$.
6	**(b)** 11	Add.
7	**(a)** 1	Count the letters with a fold line.
8	**(b)** 30	$360 \div$ spokes.
9	**(c)** 5	Count the letters with a fold line.
10	**(d)** 24	$360 \div$ spokes.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Angle between neighbouring vertices $= 360 \div 9 = 40°$ (1); Turn $= 4 \times 40 = 160°$ (1).	1+1
12	Sides $= 360 \div 24 = 15$ (1); Order $= 15$ (1).	1+1
13	regular decagon: 10 (1); regular hexagon: 6; total 16 (1).	1+1
14	Angle between neighbouring vertices $= 360 \div 10 = 36°$ (1); Turn $= 3 \times 36 = 108°$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Angle $= 360 \div 10 = 36°$ (1); After 7 turns: $252°$ (1); Turns for full rotation $= 10$ (1).	1+1+1
16	X: order 2, lines 2; Y: order 1, lines 1; Z: order 2, lines 0 (1); Compare each pair (1); Counting complete (1).	1+1+1
17	Angle between marks $= 360 \div 9 = 40°$ (1); Lines of symmetry $= 9$ for a regular arrangement (1); Turn to mark 2: $80°$ (1).	1+1+1
	OR: Angle $= 360 \div 12 = 30°$ (1); After 5 turns: $150°$ (1); Turns for full rotation $= 12$ (1).	1+1+1
18	S: order 2, lines 0; H: order 2, lines 2; O: order 2, lines 2 (1); Compare each pair (1); Counting complete (1).	1+1+1
	OR: Angle between marks $= 360 \div 15 = 24°$ (1); Lines of symmetry $= 15$ for a regular arrangement (1); Turn to mark 4: $96°$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(a) 10 (1); (b) $36°$ (1); (c) 10 (1); (d) $252°$ is 7 turns so it leaves the pattern unchanged (2).	1+1+1+2
	OR: regular hexagon: $60°$ (1); regular 12-gon: $30°$ (1); Larger angle: regular hexagon; difference $= 30°$ (1); Total lines $= 6 + 12 = 18$ (2).	1+1+1+2
20	(a) 12 (1); (b) $30°$ (1); (c) 12 (1); (d) $150°$ is 5 turns so it leaves the pattern unchanged (2).	1+1+1+2
	OR: regular pentagon: $72°$ (1); regular octagon: $45°$ (1); Larger angle: regular pentagon; difference $= 27°$ (1); Total lines $= 5 + 8 = 13$ (2).	1+1+1+2

Mistakes to avoid:

- Saying a rectangle has 4 lines of symmetry (diagonals are not lines of symmetry); a non-square rectangle has only 2.
- Giving a parallelogram 2 lines of symmetry; it has none, though it has rotational symmetry of order 2.
- Mixing up the order of rotation with the smallest angle: angle $= 360° \div$ order.
- Saying a figure with order 1 has rotational symmetry; order 1 means only the trivial full turn.
- Drawing the mirror image at the wrong distance from the mirror line or on the same side of it.

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